

HW Subsystem Test Plan

1. Power Supply, Reference Voltage and VCO Modulation Path

1.1 Test parameters

- Scope: power supply, reference voltage circuits, and optional VCO modulation input path
- Nominal PCB supply voltage: 5 V
- Nominal on-board regulated voltage: 3.3 V
- Nominal radar reference voltage: 1.65 V

1.2 Acceptance criteria

- Supply/reference checks pass if measured values are within their stated tolerances and remain stable without abnormal ripple or oscillation
- VCO modulation path checks pass if the output follows the applied input stimulus with the expected gain within the stated tolerance and without visible instability or distortion

1.3 Test plan

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
HW-1.01	3.3 V on-board regulated voltage	Power the PCB under nominal operating conditions	On-board regulated voltage = 3.3 V $\pm$ 1% and stable	3.2998 V	passed	---
HW-1.02	Current consumption	Power the PCB under nominal operating conditions with no additional modules connected (e.g. radar, OpenLog etc.)	Total current consumption $\leq$ 15 mA	9.03 mA	passed	---
HW-1.03	Radar reference voltage	Power the PCB under nominal operating conditions	Reference voltage = 1.65 V $\pm$ 16.5 mV and stable	1.6470 V	passed	---
HW-1.04	VCO_in DC transfer check	Apply a constant 1 V DC level to the VCO_in path	Output at VCO_in pin = 1.1 V DC $\pm$ 1% and stable	1.0971 V	passed	---

2. I/Q Channel Tests

2.1 Test parameters

- Scope: I and Q analog paths
- Nominal analog-path DC operating point before ADC: 1.5 V
- Intended final output operating range: 0..3 V
- Nominal in-band gain target: 60 dB
- Cut-off frequencies: 0.3 Hz and 15 Hz
- In-band gain test frequencies: 0.5, 1, 2, 5, 10 Hz
- Out-of-band attenuation check frequencies: 80 mHz and 150 Hz
- Dynamic input amplitudes: 0.5 mVpp and 2.5 mVpp
- Inactive channel condition: grounded

2.2 Acceptance criteria

- DC operating-point checks pass if both I and Q path outputs before ADC are centered at 1.5 V within the stated tolerance and show no instability
- In-band gain checks pass if measured gain remains within the stated tolerance at all defined passband points
- Cut-off checks pass if relative attenuation at the cut-off point is $-3 \text{ dB} \pm 20\%$ versus the midband response
- Out-of-band attenuation checks pass if relative attenuation at 80 mHz and 150 Hz is $-20 \text{ dB} \pm 20\%$ versus the midband response
- Waveform-quality checks pass if waveforms remain sinusoidal and unclipped at all test amplitudes
- Simultaneous I/Q checks pass if both channels satisfy the corresponding single-channel criteria under the same input condition
- Channel-matching checks pass if I/Q gain difference and phase difference do not exceed the stated tolerance across the passband
- Crosstalk / channel-isolation checks pass if the inactive grounded channel shows no unintended amplified response beyond the stated tolerance while the other channel is driven

2.3 Test plan

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
HW-2.01	DC operating point	Power the PCB under nominal operating conditions with both I and Q inputs grounded	Outputs biased at 1.5 V DC on both paths	1.47 V DC	passed	---
HW-2.02	I-path in-band gain / waveform check	Apply sine to I input only . Ground Q input . Frequencies: 0.5, 1, 2, 5, 10 Hz . Run once at 0.5 mVpp and once at 2.5 mVpp	Passband gain 60 dB $\pm$ 20% at all in-band points. Expected output $\approx$ 0.5 Vpp and 2.5 Vpp respectively, centered around 1.5 V DC . Clean sinusoidal waveform, no clipping at either amplitude	as expected, within tolerance	passed	---
HW-2.03	Q-path in-band gain / waveform check	Same as HW-2.02, with sine applied to Q input only and I input grounded	Same as HW-2.02	as expected, within tolerance	passed	---
HW-2.04	Simultaneous in-band response	Same as HW-2.02, with the same sine signal applied simultaneously to both I and Q inputs	Same as HW-2.02, on both outputs simultaneously. No instability	as expected, within tolerance	passed	---
HW-2.05	I-path cut-off check	Apply sine to I input only . Ground Q input . Frequencies: 0.3 Hz and 15 Hz . Run once at 0.5 mVpp and once at 2.5 mVpp	Attenuation -3 dB $\pm$ 20% relative to midband response at both cut-off points. Expected output $\approx$ 355 mVpp and 1.78 Vpp respectively, centered around 1.5 V DC	as expected, within tolerance	passed	---

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
HW-2.06	Q-path cut-off check	Same as HW-2.05, with sine applied to Q input only and I input grounded	Same as HW-2.05	as expected, within tolerance	passed	---
HW-2.07	Simultaneous cut-off check	Same as HW-2.05, with the same sine signal applied simultaneously to both I and Q inputs	Same as HW-2.05, on both outputs simultaneously. No instability	as expected, within tolerance	passed	---
HW-2.08	I-path out-of-band attenuation check	Apply sine to I input only . Ground Q input . Frequencies: 80 mHz and 150 Hz . Run once at 0.5 mVpp and once at 2.5 mVpp	Attenuation -20 dB $\pm$ 20% relative to midband response at both out-of-band points. Expected output $\approx$ 50 mVpp and 250 mVpp respectively, centered around 1.5 V DC	as expected, within tolerance	passed	---
HW-2.09	Q-path out-of-band attenuation check	Same as HW-2.08, with sine applied to Q input only and I input grounded	Same as HW-2.08	as expected, within tolerance	passed	---
HW-2.10	Simultaneous out-of-band attenuation check	Same as HW-2.08, with the same sine signal applied simultaneously to both I and Q inputs	Same as HW-2.08, on both outputs simultaneously. No instability	as expected, within tolerance	passed	---
HW-2.11	I/Q channel matching in passband	Apply the same sine signal simultaneously to both I and Q inputs . Frequencies: 0.5, 1, 2, 5, 10 Hz . Run once at 0.5 mVpp and once at 2.5 mVpp	I/Q gain difference $\leq$ 10% and I/Q phase difference $\leq$ 1% across the passband at both amplitudes. Both outputs remain centered around 1.5 V DC	as expected, within tolerance	passed	---

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
HW-2.12	Crosstalk / channel isolation check	Drive one channel at a time with sine while keeping the other channel input grounded (I active / Q grounded, then Q active / I grounded). Frequencies: 0.5, 1, 2, 5, 10 Hz . Run once at 0.5 mVpp and once at 2.5 mVpp	The inactive grounded path remains centered around 1.5 V DC and any coupled signal remains below 1% at all test frequencies and amplitudes	≤ 10 mVpp	passed	---

3. Radar-Connected Tests

3.1 Test parameters

- Scope: end-to-end verification of the connected K-LC5 radar module together with the I/Q analog paths
- Radar target simulator: RFbeam K-DT1
- Recommended minimum-speed setting for this section: 1 km/h (lowest available K-DT1 Doppler frequency)
- K-DT1 adjustable parameters relevant for this section: speed, direction, signal amplitude / reach, and signal duration
- Recommended mechanical setup: fix both K-DT1 and sensor board mechanically, align K-DT1 toward the radar sensor, keep spacing ≥ 30 cm, and minimize nearby parasitic reflections
- Intended final output operating range at the external amplifier outputs: 0..3 V
- Nominal analog-path DC operating point before ADC: 1.5 V

3.2 Acceptance criteria

- Radar-connected basic-response checks pass if activation of K-DT1 produces stable observable periodic signals on both I and Q outputs at the expected Doppler frequency, centered around 1.5 V DC, without clipping or instability
- Radar-connected direction checks pass if changing the K-DT1 movement direction reverses the relative I/Q phase order while keeping the Doppler frequency consistent
- Radar-connected reach / amplitude sanity checks pass if increasing the K-DT1 signal amplitude / reach increases the observed AC output amplitude correspondingly, while outputs remain within the intended 0..3 V range

Note: Since the minimum available K-DT1 Doppler frequency (44 Hz) lies above the intended analog passband, this section is treated as an integration and functionality check rather than a strict nominal-band amplitude verification

3.3 Test plan

ID	Test description	Input signals / setup	Expected result	Measured result	Verdict	Remarks
HW-3.01	Radar-connected basic response	Place K-DT1 in front of the radar sensor in a fixed aligned setup with spacing ≥ 30 cm . Set K-DT1 to 1 km/h , one fixed direction, and one representative reach setting (recommended: start with 100% , reduce only if clipping occurs)	Both I and Q outputs show stable observable periodic signals at approx. 44 Hz , centered around 1.5 V DC . Both channels remain within 0..3 V and show no clipping or instability	---	---	---
HW-3.02	Radar-connected direction check	Use the same fixed setup as in HW-3.01 . Keep speed at 1 km/h and the same reach setting. Run once with forward direction and once with backward direction on K-DT1	In both cases, observable signals remain present on I and Q at approx. 44 Hz , centered around 1.5 V DC . Reversing K-DT1 direction reverses the relative I/Q phase order. No clipping or instability	---	---	---
HW-3.03	Radar-connected reach / amplitude sanity check	Use the same fixed setup as in HW-3.01 . Keep speed at 1 km/h and one fixed direction. Compare at least two K-DT1 reach settings, for example low and high (e.g. 20% and 100%)	Higher K-DT1 reach produces a correspondingly larger observed AC amplitude on both outputs. Outputs remain centered around 1.5 V DC , stay within 0..3 V , and show no clipping or instability	---	---	---